Experiment No. 7

Aim: Implementation of different I/O operations using I/O streams

1. **Write a Java program that reads data from a sample.txt file located outside the program's directory**

```
import java.io.File;
import java.io.FileNotFoundException;
import java.util.Scanner;

public class ReadFileExample {
   public static void main(String[] args) {

      File file = new File("C:\\Users\\PRASANNA\\Desktop\\AOOC-Practicals-4th
sem\\exp7\\sample.txt");

      try {
         Scanner sc = new Scanner(file);

         while (sc.hasNextLine()) {
            System.out.println(sc.nextLine());
         }

         sc.close();
      } catch (FileNotFoundException e) {
         System.out.println("File not found!");
      }
   }
}
```

OUTPUT :

```
c:\Users\PRASANNA\Desktop\AOOC-Practicals-4th sem\exp7>cd "c:\Users\PRASANNA\Desktop
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Hello
Welcome to Java
File reading example

c:\Users\PRASANNA\Desktop\AOOC-Practicals-4th sem\exp7>
```

2. Develop a Java program that performs the following operations:
○ Accept student information such as name, age, weight, height, city, and phone number from the user.
Lab Manual: AOOC
Lab manual: Advanced Object-Oriented Concepts (Java), SY CSE, Sem IV, 2025-26, Computer Science & Engineering Dept., D.Y. Patil College of Engineering & Technology, Kasaba Bawada, Kolhapur
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○ Store this information in a file using DataOutputStream along with FileOutputStream.
○ Retrieve and display the data using DataInputStream along with FileInputStream

```
import java.io.File;
import java.io.FileNotFoundException;
import java.util.Scanner;

public class ReadFileExample {
   public static void main(String[] args) {

      File file = new File("C:\\Users\\PRASANNA\\Desktop\\AOOC-Practicals-4th
sem\\exp7\\sample.txt");

      try {
         Scanner sc = new Scanner(file);

         while (sc.hasNextLine()) {
            System.out.println(sc.nextLine());
         }

         sc.close();
      } catch (FileNotFoundException e) {
         System.out.println("File not found!");
      }
   }
}
```

OUTPUT :

```
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encodi
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encodi
Enter Name: Prasanna
Enter Age: 20
Enter Weight: 62
Enter Height: 6
Enter City: Kolhapur
Enter Phone Number: 7620809814

Data written to file successfully!

--- Student Data ---
Name: Prasanna
Age: 20
Weight: 62.0
Height: 6.0
City: Kolhapur
Phone: 7620809814

c:\Users\PRASANNA\Desktop\AOOC-Practicals-4t
```

3. Write a Java program to read a text file and compute the following:
○ The total number of vowels in the file.
○ The total number of words in the file.
○ The number of times the character 'a' appears in the file

```
import java.io.*;
import java.util.*;

public class FileAnalysis {
   public static void main(String[] args) {

      int vowelCount = 0;
      int wordCount = 0;
      int aCount = 0;

      try {
         File file = new File("sample2.txt"); // change path if needed
         Scanner sc = new Scanner(file);
```

```
            while (sc.hasNextLine()) {
                String line = sc.nextLine();

                // Count words
                String[] words = line.trim().split("\\s+");
                if (!line.trim().isEmpty()) {
                    wordCount += words.length;
                }

                // Convert line to lowercase for easy checking
                line = line.toLowerCase();

                for (int i = 0; i < line.length(); i++) {
                    char ch = line.charAt(i);

                    // Count vowels
                    if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
                        vowelCount++;
                    }

                    // Count 'a'
                    if (ch == 'a') {
                        aCount++;
                    }
                }
            }

            sc.close();

            // Display results
            System.out.println("Total vowels: " + vowelCount);
            System.out.println("Total words: " + wordCount);
            System.out.println("Occurrences of 'a': " + aCount);

        } catch (FileNotFoundException e) {
            System.out.println("File not found!");
        }
    }
}
```

OUTPUT :

sample2.txt

```
Java is amazing
And easy to learn
```

```
c:\Users\PRASANNA\Desktop\AOOC-Practicals-4th sem\exp7>cd "
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstder
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstder
Total vowels: 12
Total words: 7
Occurrences of 'a': 7

c:\Users\PRASANNA\Desktop\AOOC-Practicals-4th sem\exp7>
```

4. Write a program that takes a file name as input through the command line.
● If the file exists, open it and display its contents.
● After displaying the contents, ask the user: “Do you want to add data to the end of the file?”
● If the user's response is "Yes", accept data from the user and append it to the file.
● If the file does not exist, create a new file and allow the user to input data to store in it.
● The user should type “exit” on a new line to stop entering data. Implement this program using character stream classes

```
import java.io.*;
import java.util.Scanner;

public class FileHandler {
    public static void main(String[] args) {

        // Check command-line argument
        if (args.length == 0) {
            System.out.println("Please provide file name as argument!");
            return;
        }

        String fileName = args[0];
        File file = new File(fileName);

        Scanner sc = new Scanner(System.in);
```

```
try {
    // ----------- If file exists -----------
    if (file.exists()) {
        System.out.println("\nFile exists. Reading content:\n");

        BufferedReader br = new BufferedReader(new FileReader(file));
        String line;

        while ((line = br.readLine()) != null) {
            System.out.println(line);
        }
        br.close();

        // Ask user to append
        System.out.print("\nDo you want to add data to the end of the file? (Yes/No): ");
        String choice = sc.nextLine();

        if (choice.equalsIgnoreCase("Yes")) {
            BufferedWriter bw = new BufferedWriter(new FileWriter(file, true));

            System.out.println("Enter data (type 'exit' to stop):");

            while (true) {
                String input = sc.nextLine();
                if (input.equalsIgnoreCase("exit"))
                    break;
                bw.write(input);
                bw.newLine();
            }

            bw.close();
            System.out.println("Data appended successfully!");
        }

    }
    // ----------- If file does NOT exist -----------
    else {
        System.out.println("\nFile does not exist. Creating new file...");

        BufferedWriter bw = new BufferedWriter(new FileWriter(file));

        System.out.println("Enter data (type 'exit' to stop):");

        while (true) {
            String input = sc.nextLine();
```

```
            if (input.equalsIgnoreCase("exit"))
                break;

            bw.write(input);
            bw.newLine();
        }

        bw.close();
        System.out.println("File created and data written successfully!");
    }

    } catch (IOException e) {
        System.out.println("Error: " + e.getMessage());
    }

    sc.close();
  }
}
```

OUTPUT :

```
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstde
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstde
Please provide file name as argument!

c:\Users\PRASANNA\Desktop\AOOC-Practicals-4th sem\exp7>java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstde

File exists. Reading content:

Hello
Welcome to Java
File reading example

Do you want to add data to the end of the file? (Yes/No): y
Enter data (type 'exit' to stop):
this is experiment no. 07
exit
Data appended successfully!

c:\Users\PRASANNA\Desktop\AOOC-Practicals-4th sem\exp7>
```

sample.txt

```
Hello
Welcome to Java
File reading examplethis is experiment no. 07
```